

Buffers

- (a) Calculate the pH of a 1.0 M CH_3COOH ($K_a = 1.8 \times 10^{-5}$ M)
- (b) Calculate the pH of a 1.0 M CH_3COONa
- (c) Calculate the pH of a buffer system containing $[\text{CH}_3\text{COOH}] = 1.0$ M and $[\text{CH}_3\text{COO}^-] = 1.0$ M.
- (d) What is the pH of the solution after the addition of 0.10 mol gaseous HCl to 1.0 L of the solution (assuming no change in volume). Compare the result to the expected pH if 0.1 mol gaseous HCl is dissolved in 1.0 L pure water.

Preparing a buffer with a specific pH

The expression for the acid ionization constant reads:

$$K_a(\text{HA}) = \frac{[\text{A}^-][\text{H}_3\text{O}^+]}{[\text{HA}]} \quad \text{and therefore} \quad [\text{H}_3\text{O}^+] = \frac{K_a[\text{HA}]}{[\text{A}^-]}$$

Taking the logarithm yields

$$-\log[\text{H}_3\text{O}^+] = -\log \frac{K_a[\text{HA}]}{[\text{A}^-]} = -\log K_a - \log \frac{[\text{HA}]}{[\text{A}^-]}$$

Which is equivalent to

$$\text{pH} = \text{p}K_a + \log \frac{[\text{A}^-]}{[\text{HA}]} \quad (\text{note the fraction and sign inversion})$$

Henderson-Hasselbalch equation ("bufferligningen")

Exercise

How would you prepare a buffer with a pH of about 7.45 using "phosphates":

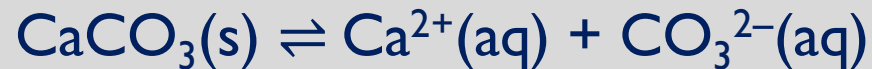


Solubility and solubility products



Solubility products

Example: Calcium carbonate is very slightly soluble in water and dissolves according to:



with the equilibrium constant (*the solubility product*)

$$K_{\text{sp}} = [\text{Ca}^{2+}][\text{CO}_3^{2-}] = 3.3 \times 10^{-9} \text{ M}^2$$

Exercise:

- Calculate the solubility (g/L) of CaCO_3 in pure water.
- In healthy coral reef areas, $[\text{CO}_3^{2-}] \approx 10^{-4} \text{ M}$ and $[\text{Ca}^{2+}] \approx 10^{-2} \text{ M}$.
Comment.
- In which direction would the above equilibrium shift during ocean acidification?

Electrochemistry

2602I Chemistry (Polytechnical Foundation)

Professor Kasper S. Pedersen

2026

Redox reactions

Redox = reduction-oxidation

Redox reaction: a chemical reaction where electrons are transferred between reactants

- **Example**, zinc in acid



Which can be decomposed into two *half-cell* reactions:



Oxidation state

Oxidation states (OS) are formal book-keeping – but they are useful for, e.g., balancing reaction schemes.

Rules for OS:

- 1) OS = 0 for free elements.
- 2) In monoatomic ions, OS = charge.
- 3) OS = $-II$ for oxygen in compounds (except O_2^{2-} and O_2^-)
- 4) OS = $+I$ for hydrogen, except when bound to metals
- 5) Halogens have OS = $-I$, except when bound to oxygen.
- 6) The sum of the OS in a molecule/ion must equal the charge of the molecule/ion.

Redox reactions – how to balance?

Conservation of electrons, atoms, and charge

Procedure:

- (1) Identify half-cell reactions
- (2) Balance the electron count by a common denominator and add half-reactions
- (3) Balance charges using H^+ in acidic solution, OH^- in basic solution (or either in neutral solution). Finally, add H_2O to balance.

Redox reactions - example

Sulfite (SO_3^{2-}) reacts with permanganate (MnO_4^-) in basic solution:



Let's balance! 🥰💃🕺

Redox reactions - examples

(I) Identify half-cell reactions



Half-cell reactions:



Redox reactions - examples

(2) Balance the electron count

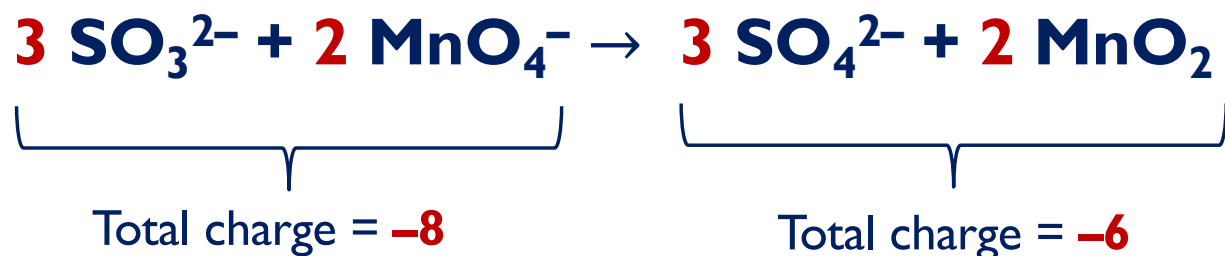


Half-cell reactions:

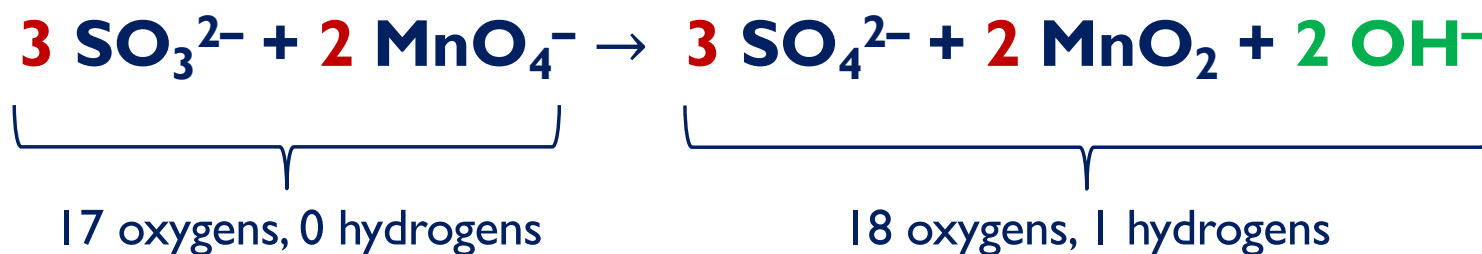


Redox reactions - examples

(3) Balance the charges



Difference of 2 in charge is compensated by adding 2 OH⁻:



Difference of oxygen and hydrogen is compensated by adding 1 H₂O:



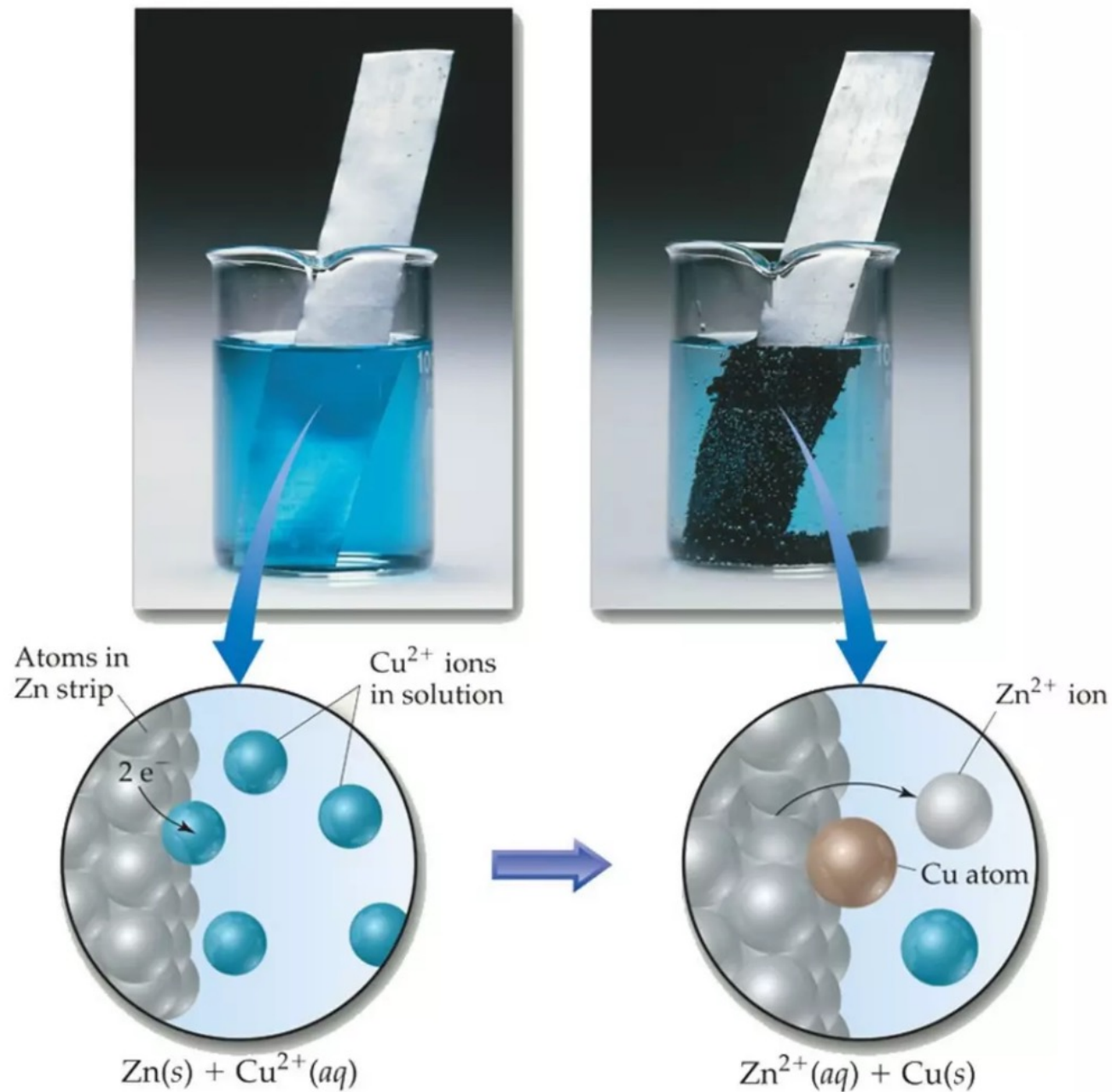
Electrochemical series

$\text{Li} \rightarrow \text{Li}^+ + \text{e}^-$	H_2 evolves on contact with cold water
$\text{K} \rightarrow \text{K}^+ + \text{e}^-$	
$\text{Ba} \rightarrow \text{Ba}^{2+} + 2\text{e}^-$	
$\text{Ca} \rightarrow \text{Ca}^{2+} + 2\text{e}^-$	
$\text{Na} \rightarrow \text{Na}^+ + \text{e}^-$	
$\text{Mg} \rightarrow \text{Mg}^{2+} + 2\text{e}^-$	H_2 evolves on contact with steam
$\text{Al} \rightarrow \text{Al}^+ + 3\text{e}^-$	
$\text{Zn} \rightarrow \text{Zn}^{2+} + 2\text{e}^-$	
$\text{Cr} \rightarrow \text{Cr}^{3+} + 3\text{e}^-$	
$\text{Fe} \rightarrow \text{Fe}^{2+} + 2\text{e}^-$	
$\text{Cd} \rightarrow \text{Cd}^{2+} + 2\text{e}^-$	H_2 evolves on contact with acids
$\text{Co} \rightarrow \text{Co}^{2+} + 2\text{e}^-$	
$\text{Ni} \rightarrow \text{Ni}^{2+} + 2\text{e}^-$	
$\text{Sn} \rightarrow \text{Sn}^{2+} + 2\text{e}^-$	
$\text{Pb} \rightarrow \text{Pb}^{2+} + 2\text{e}^-$	no H_2 generated on contact with water and acids
$\text{H}_2 \rightarrow 2\text{H}^+ + 2\text{e}^-$	
$\text{Cu} \rightarrow \text{Cu}^{2+} + 2\text{e}^-$	
$\text{Ag} \rightarrow \text{Ag}^+ + \text{e}^-$	
$\text{Hg} \rightarrow \text{Hg}^{2+} + 2\text{e}^-$	
$\text{Pt} \rightarrow \text{Pt}^{2+} + 2\text{e}^-$	
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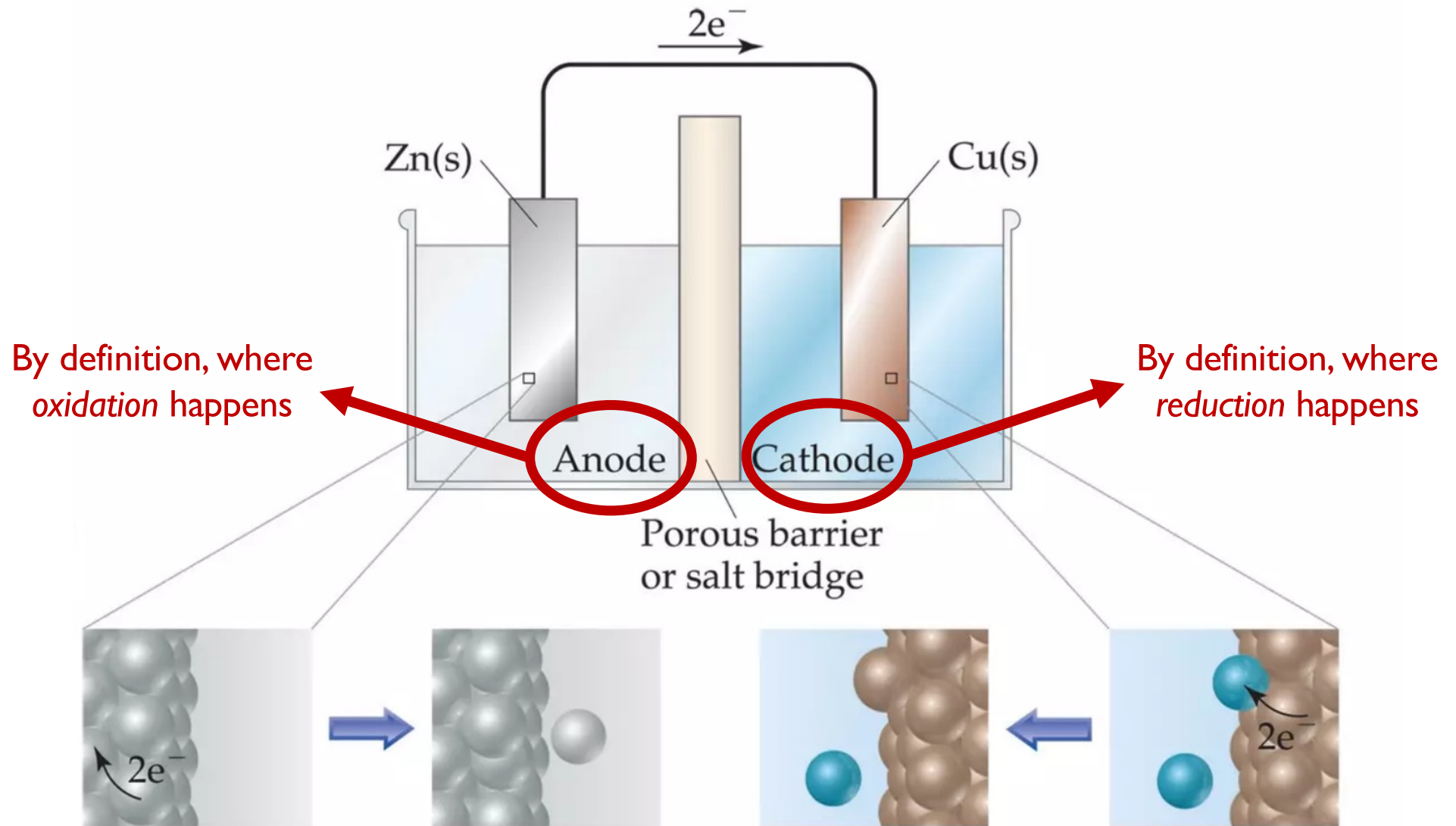
”preciousness” increases



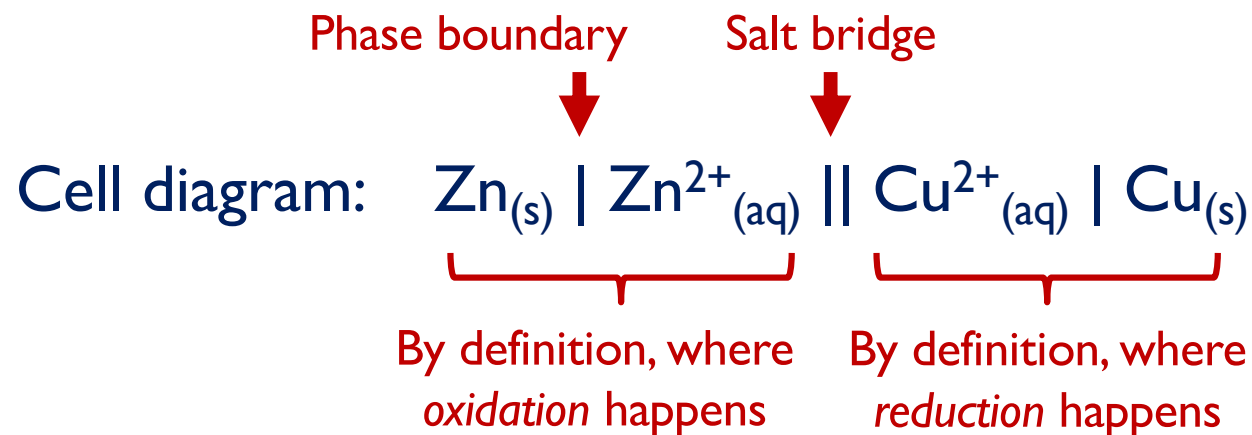
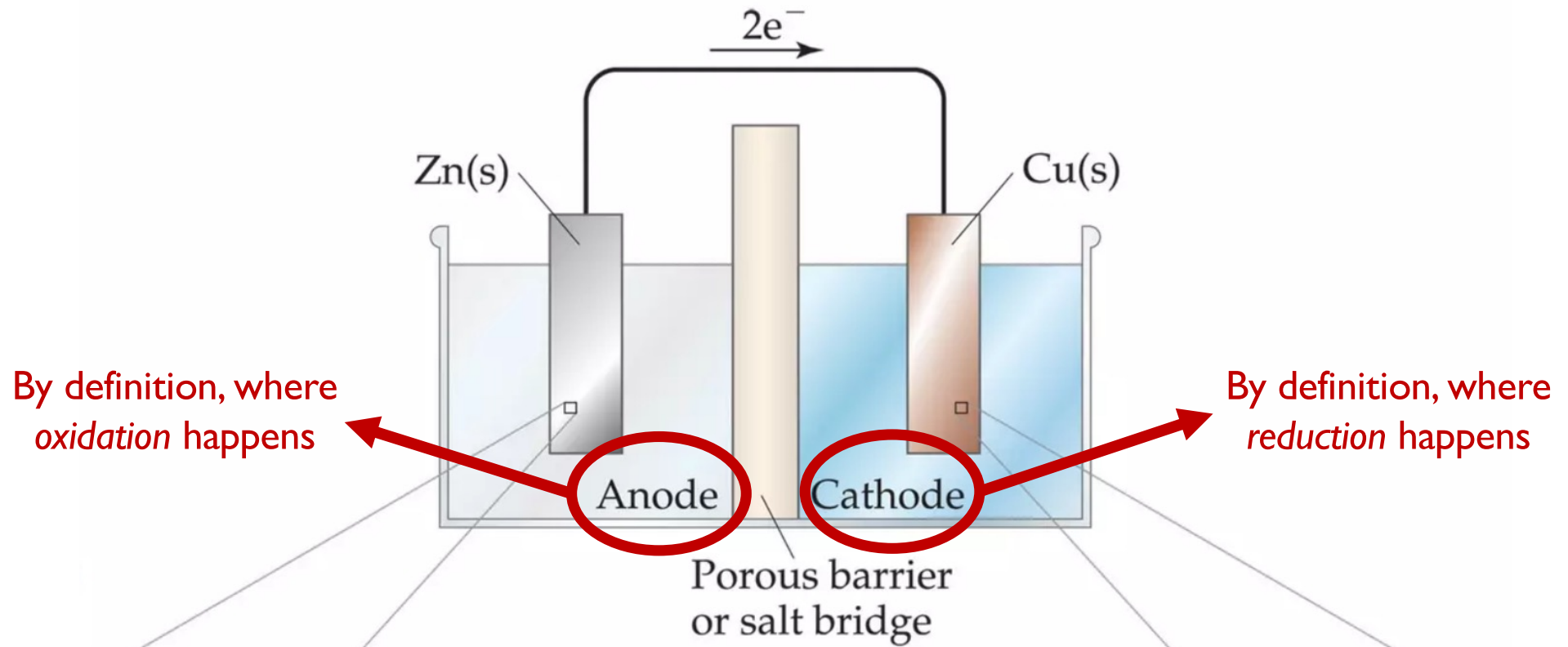
Simple redox reactions



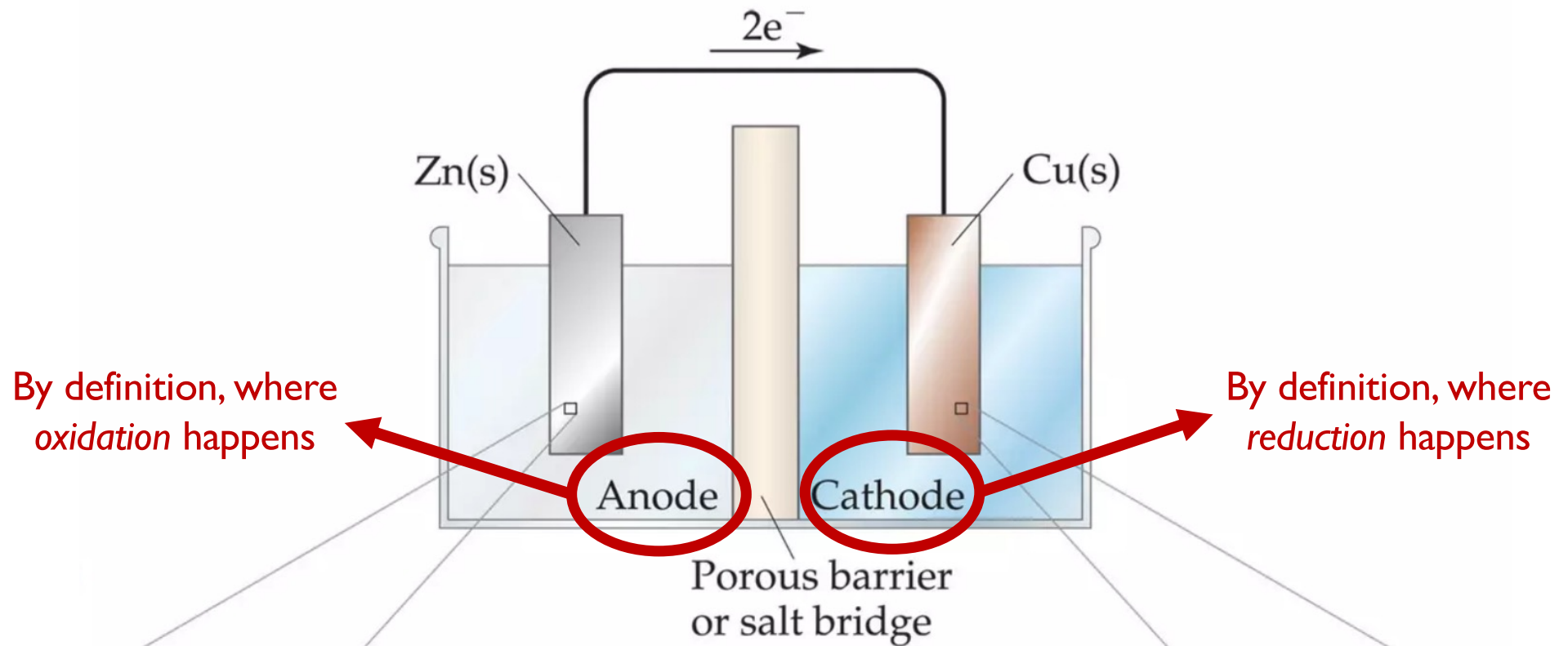
Galvanic cells – the Daniell cell



Galvanic cells – the Daniell cell



Electromotive force: E_{cell}



The measured voltage is

$$\begin{aligned} E_{\text{cell}}^{\circ} &= E_{\text{cathode}}^{\circ} - E_{\text{anode}}^{\circ} \\ &= E_{\text{red}}^{\circ}(\text{reduction}) - E_{\text{red}}^{\circ}(\text{oxidation}) \end{aligned}$$

Electromotive force is measured in volts (V):

$$1 \text{ V} = 1 \text{ J/C (i.e. "energy per electron")}$$

Standard reduction potential

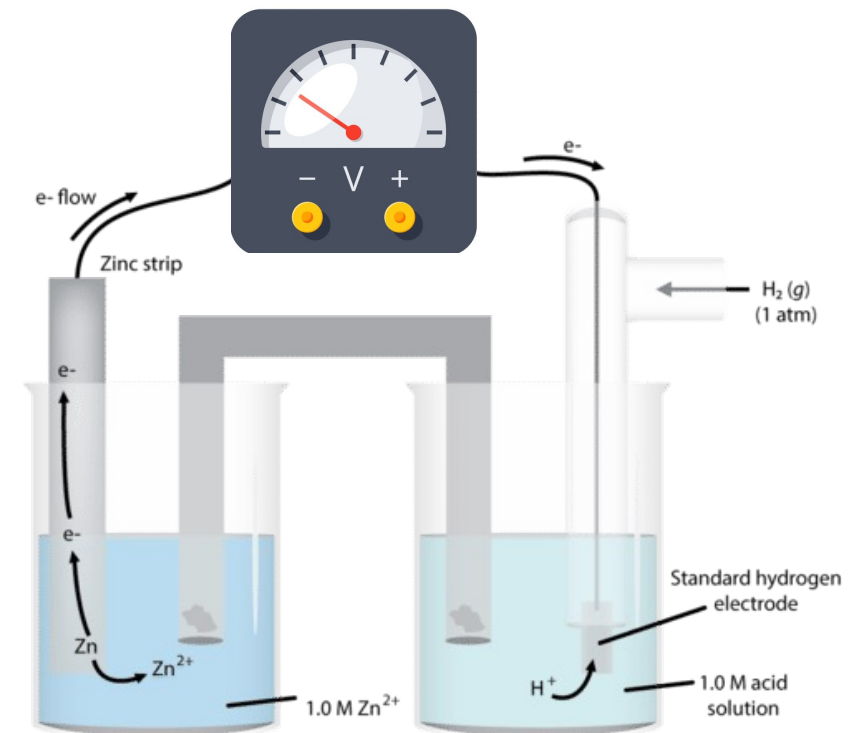
Standard hydrogen electrode (SHE)



For the half-cell reaction:



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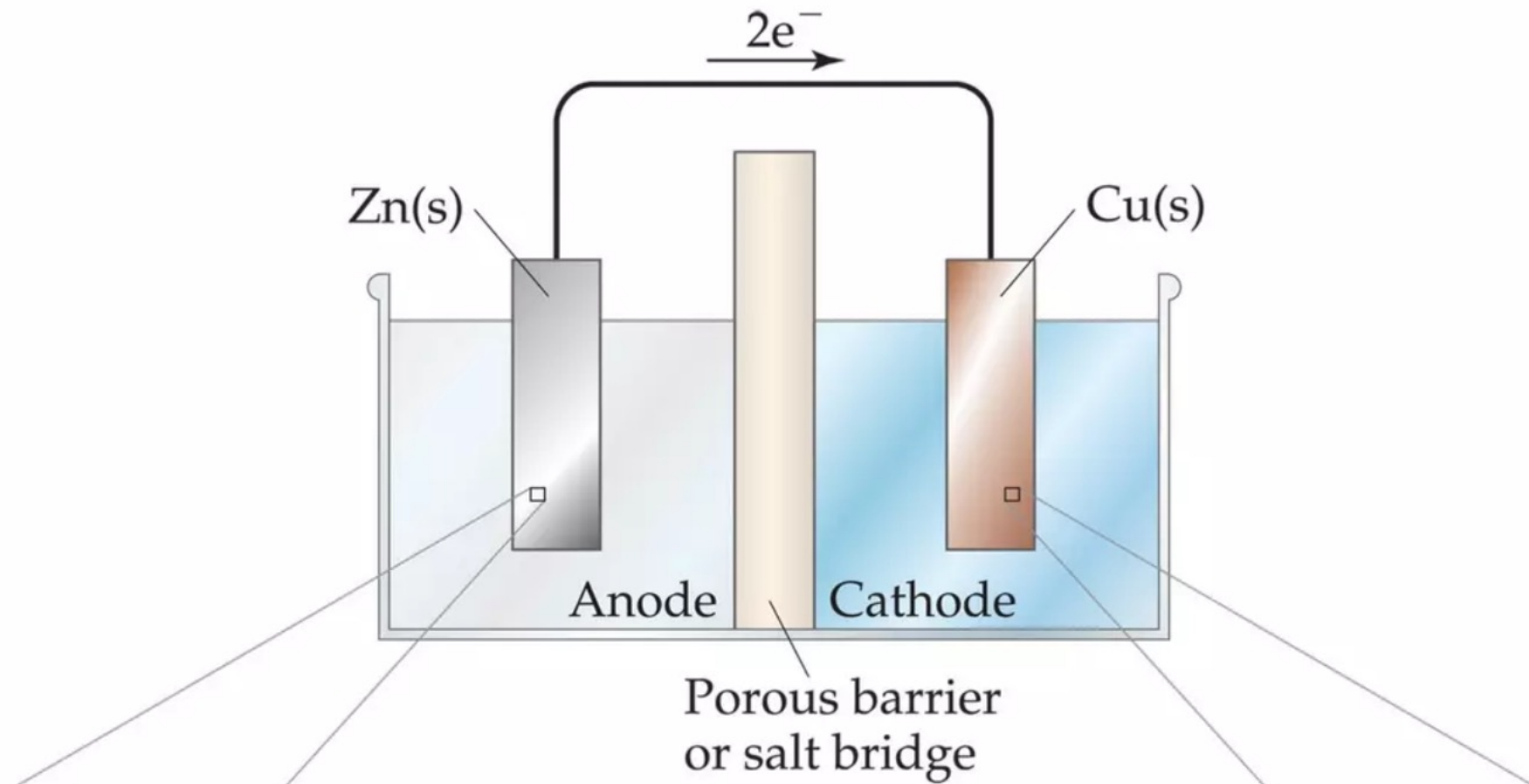


Electrochemical series

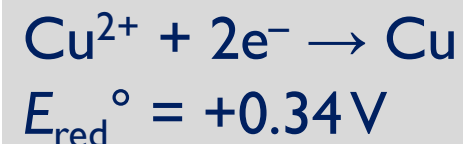
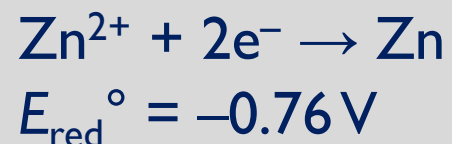
$E_{\text{red}}^{\circ} < 0$	$\text{Li} \rightarrow \text{Li}^{+} + \text{e}^{-}$	H_2 evolves on contact with cold water
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	$\text{Pt} \rightarrow \text{Pt}^{2+} + 2\text{e}^{-}$	
$E_{\text{red}}^{\circ} > 0$	$\text{Au} \rightarrow \text{Au}^{3+} + 3\text{e}^{-}$	

"preciousness" increases

Galvanic cells – the Daniell cell



Exercise: What is the electromotive force, E_{cell}° , in the Daniell cell?



Free energy of redox reactions

Electrical energy is related to voltage and charge:

$$\text{volts} \times \text{coulombs} = \text{joule}$$

The total charge of 1 mol of electrons is the **Faraday constant, F** :

$$1 F = 6.022 \times 10^{23} \text{ mol}^{-1} \times 1.602 \times 10^{-19} \text{ C} = 9.647 \times 10^4 \text{ C mol}^{-1}$$

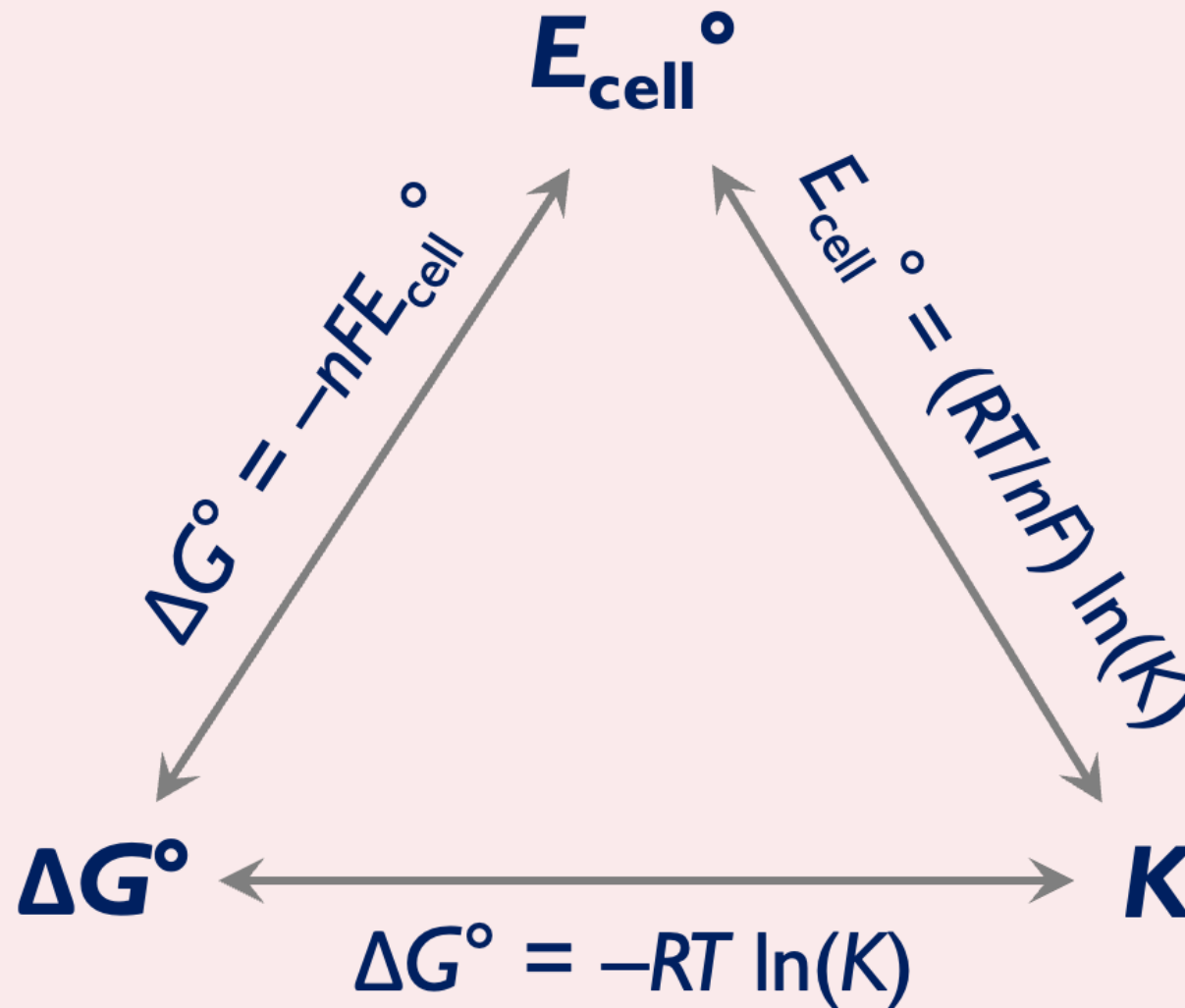
In a spontaneous electrochemical process, the electrical work done by the system is:

$$\Delta G^\circ = -n F E_{\text{cell}}^\circ$$

Recall that $\Delta G^\circ = -RT \ln(K)$, which then gives:

$$-n F E_{\text{cell}}^\circ = -RT \ln(K)$$

Free energy of redox reactions



E_{cell} vs E_{cell}° : The Nernst equation

The Gibbs energy at non-standard conditions is given by

$$\Delta G = \Delta G^{\circ} + RT \ln(Q) \quad \text{where } Q \text{ is the reaction quotient}$$

Using $\Delta G = -nFE_{\text{cell}}$ and $\Delta G^{\circ} = -nFE_{\text{cell}}^{\circ}$, we get:

$$-nFE_{\text{cell}} = -nFE_{\text{cell}}^{\circ} + RT \ln(Q)$$

Dividing through with $-nF$ yields the *Nernst equation*:

$$E_{\text{cell}} = E_{\text{cell}}^{\circ} - (RT/nF) \ln(Q)$$

A useful, simple form (at 298 K) is:

$$E_{\text{cell}} = E_{\text{cell}}^{\circ} - (0.0592 \text{ V}/n) \log(Q)$$

Galvanic cells – the Daniell cell

Exercise:

- (a) What is the change in Gibbs free energy in the Daniell cell at standard conditions?
- (b) Determine the equilibrium constant at 298 K.
- (c) What would happen to the cell potential if the concentration of Zn^{2+} is reduced to 10% of the standard conditions.

Example: Electrolysis of water

Consider the electrolysis of water:



- 1) Write out the half-cell (acidic solution):
- 2) Is the reaction spontaneous at standard conditions?

Example: K_{sp} for AgCl

An accurate way of determining solubility products is through the measurement of standard reduction potentials.

For AgCl the following potentials were measured:



Determine the solubility product of AgCl at 25 °C.